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Research

Open Food Network

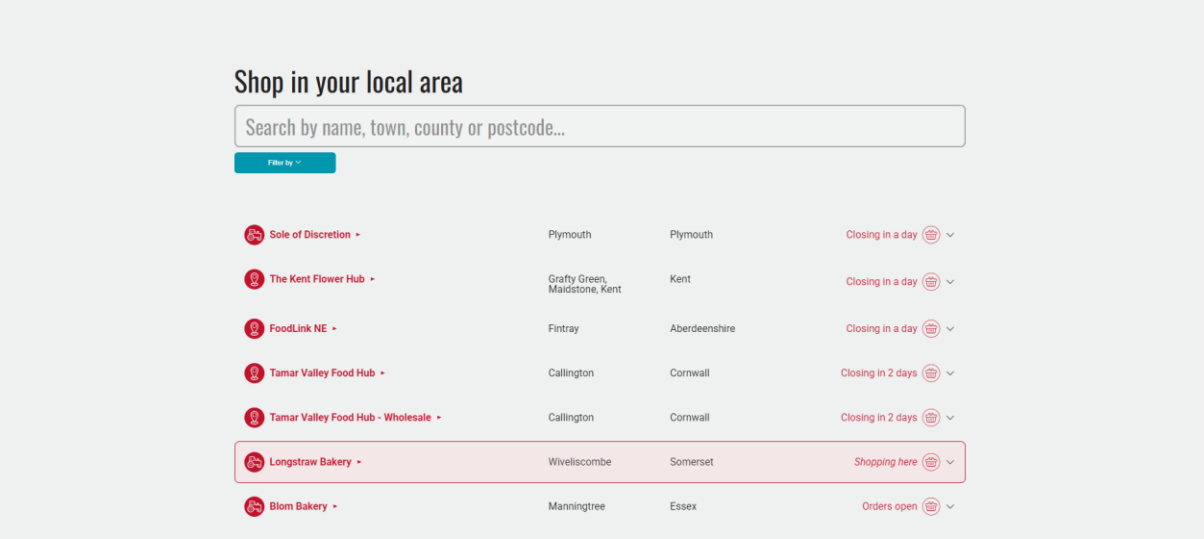
OPEN FOOD NETWORK UK

System Requirements:

This platform works on standard hardware devices like smartphones, tablets, iPads, laptops and desktops as it is cloud hosted. It requires regular and modern browsers like Google Chrome and Microsoft Edge and an internet connection. Its hardware requirements are in the medium to low range so it can run easily on most devices. It is compatible with Mobile devices, Desktop and Web browsers.

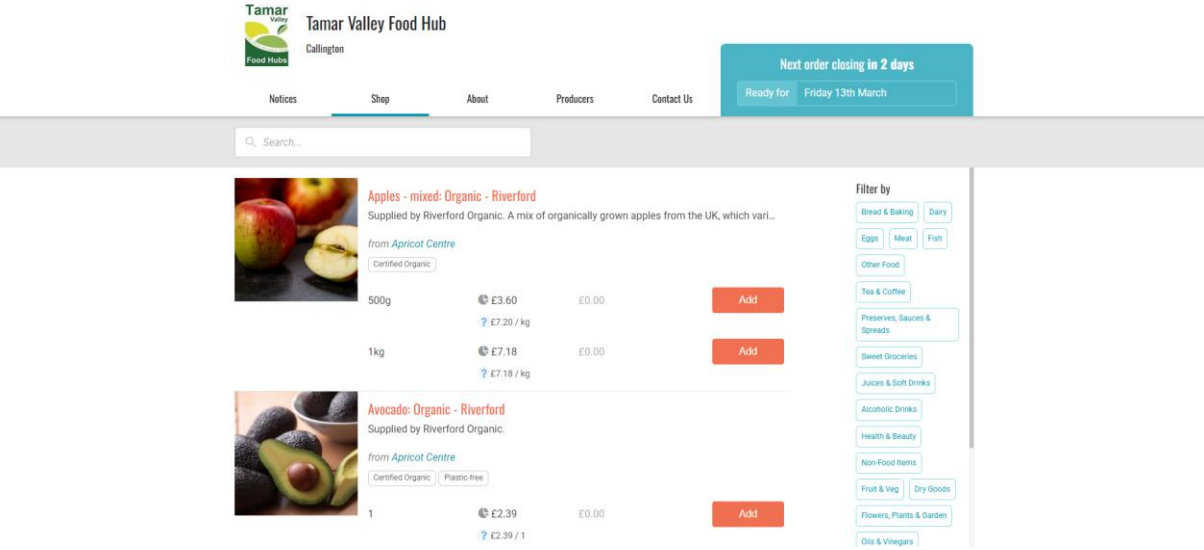
Overview:

The system connects local farmers, food places and customers whilst its features include:



- Online ordering
- Product listings with all relevant information
- Collection and delivery options
- Order management
- Inventory management

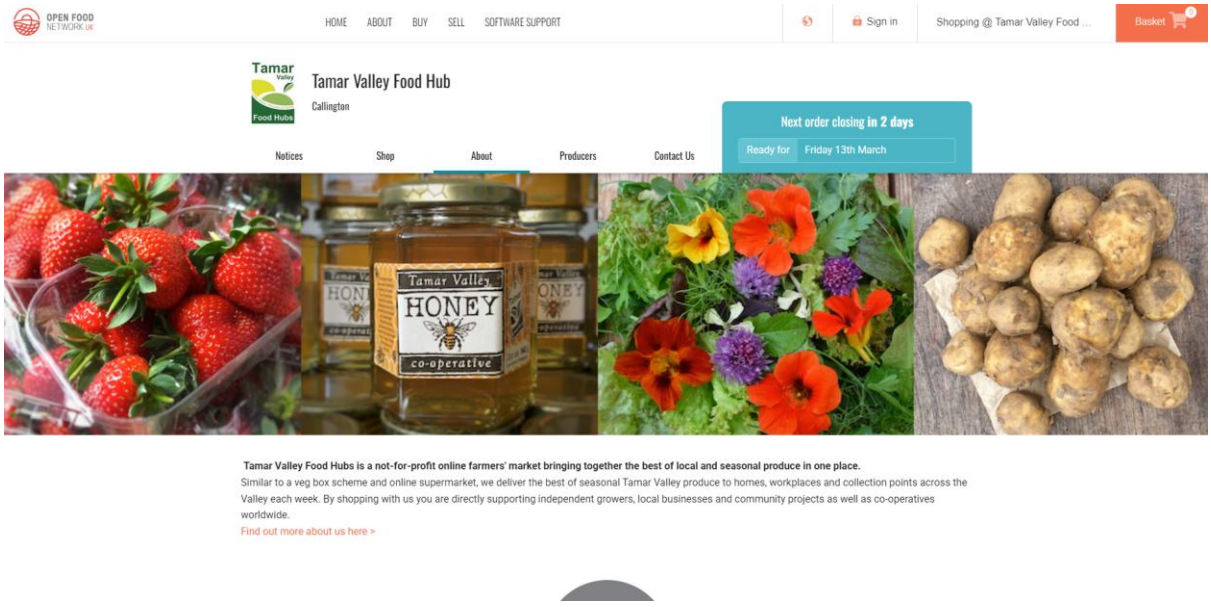
Targeting individuals like:



- Local Farmers
- Food Cooperatives
- Consumers

Which is quite industry standard and expected.

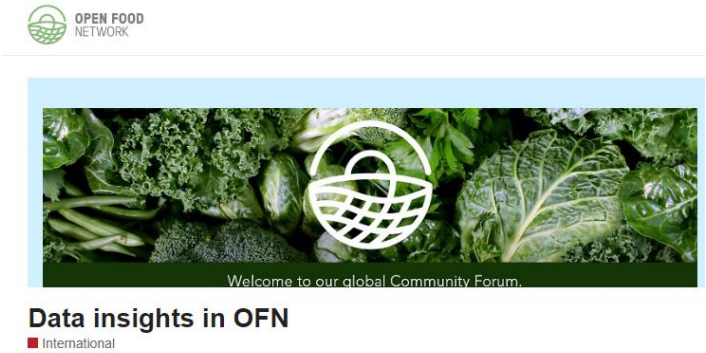
Unique Features and USP's:



- This system allows farmers to manage their own products and stock so is quite producer controlled.
- It has a plethora of producers selling on one platform.
- Customers get to see information about food, specifically where it is produced and how.

With features like these in play food producers are allowed control and maintenance over their own products while customers get to receive clear and specific information straight from the producers thus increases transparency, trust and fosters genuine customer relationships on the platform.

The platform uses new technologies like:



- API integrations.
- Data analytics for sales

Which are now common in such industries that require reliable performance and systems that are scalable.

User Interface and Accessibility:

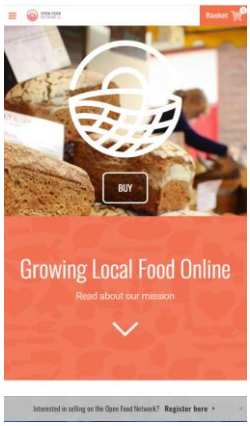
The interface is quite clean and modern and makes use:

- Clean layouts
- Natural and vibrant colours

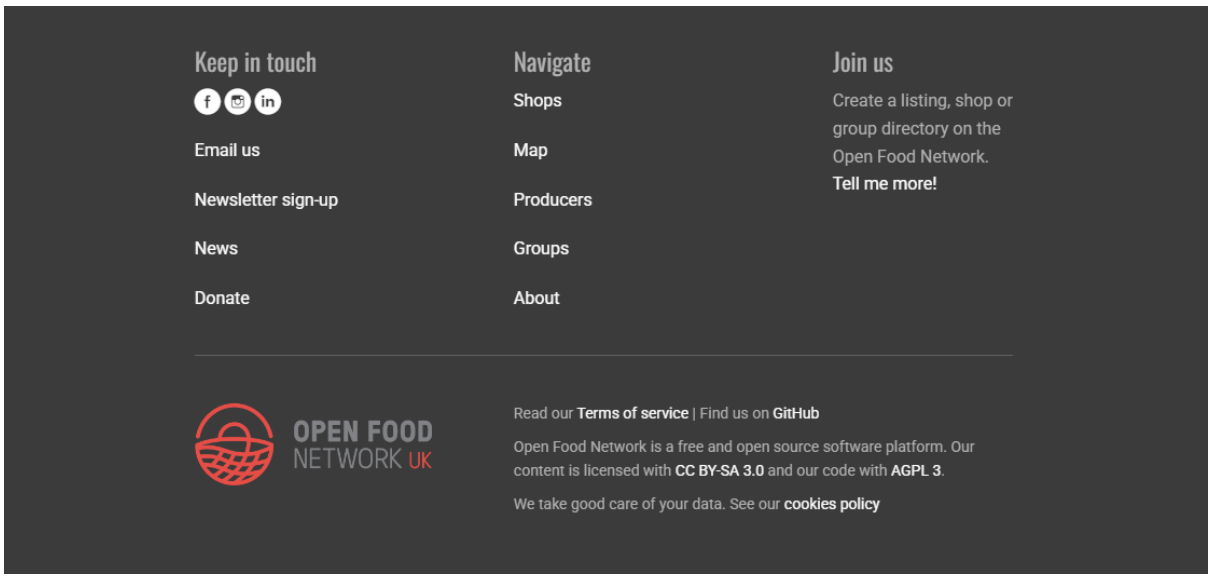
- HD image placements
- Simple navigation menus visible.

The fonts are readable and language used is straight-forward and consistent. The first impression gets the main mission and message through making it extremely effective and consumer friendly.

Accessibility Features include:

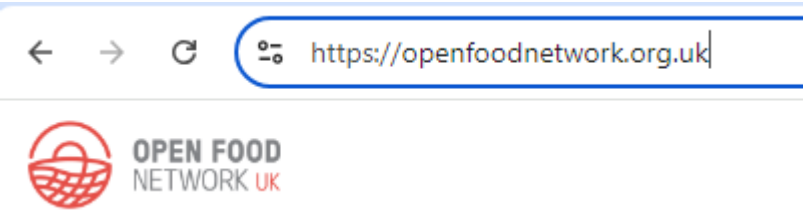


- Clear navigation menu at the top which turns into hamburger menu on mobile devices.
- Mobile responsive layout
- Alternative text for images.



Whilst it also includes help tools and help guides at the bottom for users like “Tell me more!”.

Security and Legal:



Things this system does well is making sure account login systems are secure, the payment processing is safe and secure and overall the system runs on an https connection so it is an encrypted one. Furthermore, This system ensures it adheres to legal factors including :

- Consent to users for cookies.
- It follows GDPR regulations
- It has Copyrights or permission for an images used.

Too Good To Go

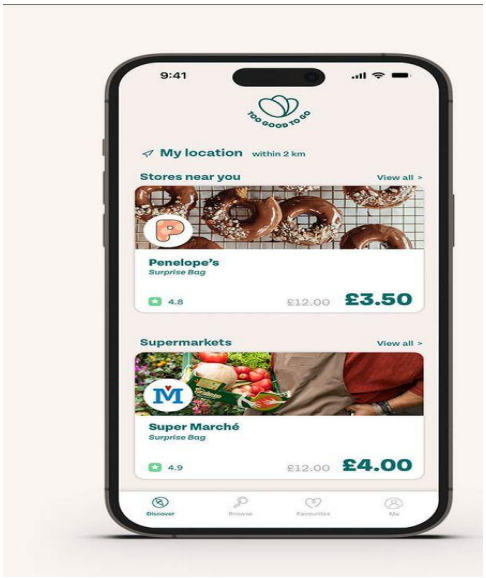
Unlike Open Food Network who focus on producer-controlled marketplaces, Too Good To Go emphasises sustainability through waste reduction of food.

This is not strictly a farmer marketplace but provides ordering, collection and discounts which are industry standard systems.

System Requirements:

This application requires smartphone device and an internet connection, The hardware required is quite low and does not need to be anything fancy.

Features Include:



- Food listings of shops nearby
- Discounted food offers at specific times
- Order reservations where you come collect
- Collection time scheduling or the application tells you when collection is open or time range

The main target audience is environmentally friendly customers and individuals looking for discounted food.

Unique Features and USP's:

- It prevents food wasting as businesses are selling unsold food instead of wasting it.
- It is based on location so uses a GPS system to pin-point establishments near the user.

This makes it a very unique business model and encourages sustainable purchasing behaviours for customers.

Technologies used include:

- GPS System
- Mobile payment systems in place.
- Data analytics.

Making it an extremely efficient system.

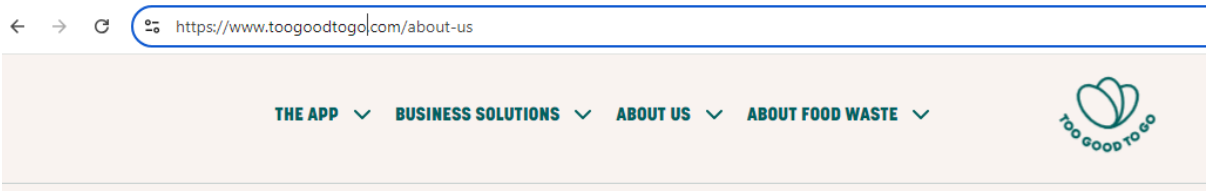
User Interface and Accessibility:

- Nice calming colours and different shades of green, cream and orange.
- Large and vibrant product cards.
- Simple and understandable navigation.

The interface is designed for quick ordering or pointing consumers exactly where they need to go.

Accessibility for this system includes its straight forward navigation, the readable fonts available and clear icons and headings there. Nevertheless, accessibility for those who are visually impaired can be improved further by adding text to speech systems.

Security and Legal:



The system ensures security by using user accounts and secure payments processing methods. It runs on an https so is encrypted ensuring safe data transfer.

For this system legal compliance would require compliance to the GDPR for safety of the consumers and the business itself, privacy policy and terms of service options available.

Business Context

The problems being solved by Greenfield Local Hub (GLH) have varying reasons behind them.

Provide customers with information about the producers, their methods and the benefits of buying locally

The problem being solved here is the lack of transparency between the food producers and customers within the traditional supply chain system. Customers would often purchase food from supermarkets when doing their groceries with no real knowledge regarding to where the product originates, how exactly the product is produced and if the practices at that location are even sustainable. This lack of information can cause curious customers to do their own research and when they find their desired results it in turn reduces customer trust in the business, likewise it also disconnects local communities from the various aspects of agriculture. Providing a system that ensures information about the producers, their methods used and the overall benefits of buying locally allows customers to make more informed purchasing decisions whilst strengthening the relationship between producers and the community around them. This benefits the business by increasing customer trust through transparency whilst increases the public image and reputation of GLH as it will be viewed as a business that prioritises sustainability and ethical food production thus increasing revenue and having a strong customer to business relationship leading to a high level of customer retention.

Allow customers to view products with transparent pricing and availability

The problem being solved here is the limited or hidden visibility customers often face regarding to pricing and availability of produced foods. Without a centralised digital platform customers may often struggle to know which producers are selling specific food items they need and if that product is available and how much it will cost. Ensuring that customers can browse for products with clear descriptions regarding to availability and

pricing reduces slow purchasing decisions for customers and reduces the risk of causing the customers to be frustrated and moving to competitor systems instead. It also ensures efficiency as customers do not need to contact producers individually for information. This benefits the business massively as it increases potential sales due to products from multiple producers being available on one single platform with all the information needed available.

Allow customers to place orders for collection or delivery

The problem being solved here is the inconvenience associated with purchasing directly from multiple local producers. Customers may need to visit multiple pages, click through multiple links and communicate through multiple informal channels even some cases where they coordinate collection manually which discourages many customers from buying locally. Having an online system that supports both delivery and collection streamlines the entire process for customers thus this modernises the way local produce is sold ensuring the business is competitive within the current market as these are industry standard practices. Subsequently, this also simplifies the order management and accessibility for customers who may not have the time to individually visit producers.

Allow customers to manage their accounts and view order history

The problem being solved here is the lack of a structures customer management system when purchases are made through informal systems such an email or phone calls. This then makes it difficult for customers to easily track their previous purchases and even manage their personal information so introducing secure user accounts allows customers to manage their details and track their previous purchases and history. This makes it very convenient for returning customers whilst it benefits the business massively as they can get valuable insight into purchasing trends and customer preferences so can recommend those products higher. This data also can help the business make calculated marketing campaigns, loyalty schemes and better inventory planning. Overall, this helps grow the business's customer base.

Provide a dashboard to enable producers to manage stock levels, update product details and view orders

The problem being solved here is the difficulty producers may face when managing all their orders and inventory without the use of digital tools. Manual and outdated processes like that of spreadsheets or paper records could lead to possible misinformation or missing information thus allowing for a dashboard where the farmers can update their product listings, manage all their stock and review incoming orders in real time ensures more accurate and up to date information is available and reduces the risk of overselling items that are out of stock. This in turn improves the overall efficiency of the platform and

empowers producers to manage their own products alone. Scalability of the system is also allowed as more producers join the system.

Order management, scheduling and tracking features

The problem being solved here is the lack of coordination that can take place when multiple customers and producers are involved in the ordering, Deliveries and collections can become extremely disorganised without structured order management tools thus leading to potential missed orders and unhappy customers. Implementing a system that tracks orders and even scheduling features ensures that both customers and producers can see the order status, its collection times or delivery schedules thus improving communication and efficiency of the system. This benefits the business by supporting operational organisation ensuring GLH maintains a service that is industry standard and extremely reliable for customers.

A loyalty scheme that provides discounts and special offers

The problem being solved here is retaining the customers as without an incentive they can continue shopping at larger and more convenient supermarkets. Putting in place a loyalty scheme that will reward repeat purchases for customers or gain points for special promotions over time makes the customers feel valued and rewarded by the business thus fostering good customer relations and increased loyalty to the business over time. Subsequently, this ensures the business is maximising as much profit as possible due to a loyal customer base whilst it encourages repeat purchases which assists local producers as well thus resulting in them praising the platform as well.

Accessibility features to support a wide range of users

The problem being solved here is compliance as the digital platform wants to ensure it complies to standards that allow individuals with different needs to navigate and use the service to its full potential. Prioritising the accessibility enhances the business's brand reputation and even the individuals they will be able to reach. This overall improves user experience for all customers thus creating an inclusive digital environment and building long term customer loyalty and demonstrating a strong sense of their corporate social responsibility which would reach and connect with the modern day audience.

Project Aim

The aim of the project is to provide the client with a fully responsive web-based application for Greenfield Local Hub (GLH) that modernises the way that cooperatives connect local

farmers and food producers with customers. This platform will streamline processes on how products are showcased, ordered and managed while promoting the overall benefits of buying locally produced food. Subsequently, because it is a centralised digital system it will improve transparency and simplify the order management system as a whole. To ensure the solution meets these objectives the system will need to:

I. Provide customers with information about the producers, their methods and the benefits of buying locally.
II. Allow customers to view products with transparent pricing and availability.
III. Allow customers to place orders for collection or delivery.
IV. Allow customers to manage their accounts and view order history.
V. Provide a dashboard to enable producers to manage stock levels, update product details and view orders.
VI. Support order management, scheduling and tracking features.
VII. Have a loyalty scheme that provides discounts and special offers.
VIII. Have accessibility features to support a wide range of users.

Functional Requirements

I. Allow users to create and manage a customer account using a secure email and password.
II. Allow customers to browse products from many local producers with in depth descriptions.
III. Allow customers to view product pricing, availability and the information relating to the producer.
IV. Allow customers to add products to a shopping cart.
V. Allow customers to place orders for collection or delivery.
VI. Allow customers to view and manage their order history.
VII. Provide producers with a dashboard to manage their products and stock levels.
VIII. Allow producers to update product details like pricing, descriptions and availability.
IX. Allow producers to view and manage incoming customer orders.
X. Allow administrators to manage users, producers and system content.
XI. Provide a loyalty scheme that rewards repeat customers with discounts or special offers.
XII. Provide an order management and tracking feature for customers and producers.
XIII. Display a message of acknowledgement or confirmation message regarding orders and updates to customers.
XIV. Include accessibility features to support users with different needs.

Rodney Blue – New Customer

Samantha Riverdale – Registered (Returning) Customer

Sylvia Anastas – Farmer / Producer

Munya Thomas – Administrator / GLH Manager

The following table presents the functional requirements alongside user stories, acceptance criteria and justifications to show how the proposed system will meet the needs and demands of different stakeholders.

Functional Requirement	**User Story**	**User Acceptance Criteria**	**Justification**
1. Allow users to create and manage a customer account using a secure email and password.	As Rodney Blue, I want to create a secure account using my email and password so that I can place orders and manage my personal details on the platform.	1. The registration page must include email and password input options. 2. System must validate email and password format. 3. Upon successful registration, a user account must be created and accessible when logged in.	Allowing the customers to create accounts solves the problem of fragmented purchasing processes where the customer would usually repeatedly enter details when ordering products. By implementing such a system the users can store their orders efficiently and this would benefit the business as this now allows GLH to gather needed customer insights which can be used for improving services and supporting future marketing strategies thus increasing the probability of acquiring potential new customers.
2. Allow customers to browse products from many local producers with in depth descriptions.	As Rodney Blue, I want to browse products from different producers so that I can see what local food items are available to purchase and I can make informed decisions after reading descriptions.	1. The system must display a list of available products. 2. Each product must show information including producer name, description and price.	Traditional food supply chains experience a lack of transparency due to no product browsing features and no in-depth descriptions. The business adding such a feature allows them to tackle this issue and allows for customers to know what local produce is actually available and who is producing it, This increases visibility for the producers who are using the system as

		3. Products must include images where available. 4. Users must be able to navigate product listings easily.	customers can browse from multiple producers while it allows customers to easily discover and purchase local food increasing the potential of revenue for the business.
3. Allow customers to view product pricing, availability and the information relating to the producer.	As Rodney Blue, I want to view product prices, availability and information about the producer so that I can make informed purchasing decisions.	1. Product listing must display product price, availability and producer name. 2. Each product must include description and producer information. 3. Information must be clearly displayed on the relevant page.	This solves the problem of lack of transparency within the food supply chain as customers are now able to know and understand where their food is coming from, how much it costs before they make any purchasing decisions thus building trust between customers and producers whilst helping them make more informed purchasing decisions overall increasing customer confidence in the system.
3b. Allow customers to view product pricing, availability and the information relating to the producer.	As Sylvia Anastas, I want to allow for customers to see accurate and reliable information about my products and farming methods so that they can know and understand where the food comes from and feel confident buying them from me.	1. Producers must be able to include information about the methods they use for production and their farm. 2. Product page must display producer information clearly. 3. Customers must be able to view producer details alongside the	This tackles the problem of transparency as the system would strengthen the connection between producers and customers since they are allowing them to showcase their information. Such transparency increases customer trust in the platform as they understand origin and how the business is ensuring sustainable food processes. This would benefit even the producers as they are able to promote their farming practices and show their own unique selling points as compared to competitors and larger retailer. This benefits the

		product listing.	business by creating a platform where both customers and producers enjoy using it thus increasing positive public image of the business and in turn users of the system as well.
4. Allow customers to add products to a shopping cart.	As Samantha Riverdale, I want to add my orders to a shopping cart so that I can purchase multiple products in a single order.	1. Users must be able to add products to a cart. 2. The cart must display selected products and quantity. 3. Users must be able to update quantities or remove items. 4. The cart must display the total price of the order.	Having such a feature in place solves the problem of a frustrating and inefficient shopping experience for users, having it improves the overall user experience when using the system allowing the user to select multiple products before adding them to a shopping cart. Without this feature in place, users would have to purchase items one after another, creating a discouraging shopping experience for users which could result in the loss of customers as they may flee to competitor applications as this feature is quite industry standard and expected. Having this feature allows the business to streamline the purchasing experience for users and ensuring the user is satisfied thus in turn increasing the customer retention for the business.
5. Allow customers to place orders for collection or delivery.	As Samantha Riverdale, I want to choose between delivery or collection when I place an order so that I can receive my items in the most	1. Checkout process must allow choice between delivery or collection. 2. Customers must be able to review their orders before submission.	Providing flexible ordering options for customers directly tackles the logistical issues that customers often face when ordering products from multiple local producers. Allowing customers to choose results in them choosing options that are more convenient to them thus encouraging them

	convenient way at that time.	3. A confirmation must be displayed to the user that the order has been made successfully.	to purchase more food from these producers as they do not worry about accessibility or convenience thus encouraging not only the customers to keep purchasing but the local producers to keep using the system as well and promoting it which ensures the public image of the business remains positive and the business model runs smoothly and efficiently.
6. Allow customers to view and manage their order history.	As Samantha Riverdale, I want to view and manage my order history so that I can track previous purchases and manage my orders if the need presents itself.	1. The system must provide an order history section within the customer account dashboard. 2. Previous orders must display details including products, quantities and the dates that the orders were placed. 3. Users must be able to view the status of the orders.	Putting this feature in place solves the problem of lack of transparency for customers in regards to what they have purchased previously, allowing it ensures transparency for customers as they can view their previous purchases. This makes it easier for them to reorder the same items they usually order which benefits the business as they can generate more profit due to convenience for customers to order. Subsequently, The business can use this data to see what customers enjoy purchasing the most and use the findings for marketing purposes thus improving the system and how it is used by customers.
7. Provide producers with a dashboard to manage their products and stock levels.	As Sylvia Anastas, I want to manage my products and stock levels through a dashboard so that customers can always see accurate	1. Producers must be able to log into a producer dashboard. 2. Producers must be able to add new products.	Providing producers with their own dashboard solves the problem of inaccurate and outdated product information being available when these details are handled manually or by individuals who are not the producers. This feature thus empowers producers to

	product availability whilst it makes it easier to keep track of my inventory.	3. Producers must be able to update product descriptions, prices and stock levels. 4. The change producers make must update in the system immediately.	manage their own listings and benefits the business as customers do not have to worry about inaccurate product listings or information and in the event of such because all information is available, they can consult the producer for assistance on their specific products. In the same manner, this allows for the administrator working on the system to focus on other bigger issues like the functionality and maintenance of the system queries rather than product listing issues ensuring smooth flow of operations.
8. Allow producers to update product details like pricing, descriptions and availability.	As Sylvia Anastas, I want to update product details like pricing, descriptions and the availability so that the information presented to customers is accurate and there is full transparency.	1. Producers must be able to edit product information from their dashboard. 2. Producers must be able to update prices, descriptions and the availability of a specific product. 3. Updates must immediately reflect on the product listings.	This feature solves the problem of inaccurate information being displayed to customers, allowing the producers access to such a system makes certain that customer see up to date information thus reducing risk of customers ordering unavailable products while allowing producers to maintain control over their listings. This benefits the business by ensuring positive customer trust with good operational efficiency as data is accurate.
9. Allow producers to view and manage incoming customer orders.	As Sylvia Anastas, I want to view orders that include my products so that I can prepare	1. Producers must be able to view orders that contain their products.	Having this feature in place directly tackles the problem of lack of coordination between the customers and producers and confusion or misinformation. Allowing

	them for collection or delivery.	2. Orders must display order quantity, status and product details. 3. Producers must be able to update the status of orders.	this feature improves the overall coordination of operations. It ensures that the products are prepared at a rate that the producer can manage, allowing for a system where customers are kept informed about the progress of their orders thus putting in place a system where customers do not have to wait too long for their products and get frustrated or producers do not have to burden themselves with too much to handle improving overall efficiency. This would then benefit the business by having a system where both customers and producers are satisfied therefore prompting for a higher user retention and in turn allowing and opportunity for the business to generate higher profits.
10 Allow administrators to manage users, producers and system content.	As Munya Thomas, I want to manage users, producers and system content so that the platform remains organised and secure.	1. Administrators must be able to view all registered users. 2. Administrators must be able to approve or remove producer accounts. 3. Administrators must be able to update system information when necessary.	Allowing this feature ensures the system to be organised and secure by allowing authorised staff to oversee user account, producer accounts and all the system content produced. This helps maintain integrity on the platform whilst ensuring it is efficient thus improving overall satisfaction of all users and on the platform thus boosting loyalty and increasing customer lifetime value.
11. Provide a loyalty scheme that	As Samantha Riverdale, I want	1. The system must record	This solves the problem of long term or regular

rewards repeat customers with discounts or special offers.	to be able to receive rewards when I make orders so that I feel encouraged to keep using the platform.	customer orders. 2. Customers must accumulate loyalty rewards based on orders. 3. Customers must be able to view the rewards or loyalty schemes available in their account.	customers feeling undervalued or unnoticed, A loyalty scheme encourages repeat purchases and would improve customer retention. By rewarding their customers, the business can build stronger long term relationships while maintain high and consistent operations.
11b. Provide a loyalty scheme that rewards repeat customers with discounts or special offers.	As Munya Thomas, I want to manage loyalty rewards and promotional offers so that the platform can encourage repeat purchases and maintain good customer engagement.	1. Administrators must be able to create or update loyalty reward offers. 2. The system must allow administrators to set discount values and the reward promotions. 3. Depending on the customer's purchase history they must automatically receive rewards.	Having this feature in place for producers tackles the problem of lack of flexibility of the system as the ability to manage loyalty rewards ensures the system remains versatile and responsive to the business needs. Since promotional offers can be adjusted when necessary the business can encourage repeat purchases whilst maintain good customer engagement which benefits the business by supporting long term business growth simultaneously as the system remains productive.
12. Provide an order management and tracking feature for customers and producers.	As Samantha Riverdale, I want to track the status of my order so that I know when it is ready to be collected or delivered.	1. Customers must be able to view the current status of their order. 2. Order statuses such as "Pending" or "Ready for Collection"	This solves the problem of lack of communication between producers and customers, having in place such a system improves the transparency of operations and ensures customers are able to stay informed about their orders while the producers can manage and

		must be displayed. 3. The order status must update when the producer changes it.	handle their schedules as efficiently as possible thus helping the business by reduced anxiety from customers as they can see the progress of their orders and good customer retention as there would be trust between the business and customers.
12b. Provide an order management and tracking feature for customers and producers.	As Sylvia Anastas, I want to update order statuses so that customers know when their order is ready.	1. Producers must be able to update order status from their dashboard. 2. When the producer updates an order, it must reflect to the customer in their account area.	This solves the problem of bad operational efficiency as allowing producers to update order statuses would improve the efficiency and coordination of all operations and would ensure customers receive accurate updates about purchases thus increasing customer trust and increased lifetime value.
12c. Provide an order management and tracking feature for customers and producers.	As Munya Thomas, I want to monitor orders placed through the platform so that I can ensure everything within the system operates as expected and any unexpected issues are resolved.	1. Administrators must be able to view a list of all orders placed on the platform. 2. Each order must have relevant information including both customer and producer details, the products ordered and their status. 3. Order updates made by a producer should be viewable by	This solves the problem of a lack of monitoring. Providing such visibility to administrators allows for the business to monitor platform activity and ensure operations and transactions are running smoothly. This in turn enables staff to quickly attend to any disputes or fix any technical issues while maintaining a very high level of service quality. This benefits the business by ensuring proactive conflict resolutions so any disputes would be sorted and handled quickly and operational quality control as there is real time monitoring so any bottlenecks identified will be quickly attended to.

		the administrator. 4. In the event of a dispute the system must allow administrators to look into it and investigate.	
13. Display a message of acknowledgement or confirmation message regarding orders and updates to customers.	As Rodney Blue, I want to receive a confirmation message when I place an order so that I know it has gone through and was submitted successfully.	1. System displays an order confirmation message after checkout. 2. Customers must receive messages when order status has been changed. 3. Confirmation message must include order details.	This solves the problem of uncertainty in customers after making an order, a message displaying makes for improved communication between the system and users. It makes customers aware that their order has been processed successfully thus benefitting the business by customers building trust in the system so will use it more often.
14. Include accessibility features to support users with different needs.	As Rodney Blue, I want the platform to be accessible so that I can navigate and use the available systems regardless of my accessibility needs.	1. The system must support keyboard only navigation. 2. Images must use alternative text. 3. The User interface must follow accessible colour contrast guidelines. 4. The platform must be usable across different devices.	This solves the problem of a compliance risk to the business by not meeting mandatory standards like WCAG. Having these in place would reduce market exclusion and ensure the platform can be used by a wide range of users including users with disabilities. Implementing these features will improve usability of the overall system while ensuring compliance. This would benefit the business by reducing the risk of any lawsuits or regulatory fines, whilst prompting for market expansion as the business would be capturing a

			demographic that competitors may be ignoring with increased accessibility functionality.
14b.Include accessibility features to support users with different needs.	As Munya Thomas, I want to see the system be accessible to all users so that Greenfield Local Hub complies to accessibility standards and reaches a wider audience.	1. The system must support keyboard only navigation. 2. Images used must have an alt text. 3. Text and colour contrast used must meet the WCAG.	Including accessibility features would address compliance and inclusivity issues that could exclude users with disability. Ensuring the system meets accessibility standards helps the business so that it can serve a wider audience thus enhancing brand reputation whilst fulfilling corporate social responsibility which is increasingly valued by modern users and is becoming quite expected within the industry.

Non-Functional Requirements

The following non-functional requirements define the characterisitcs that the system must demonstrate to uphold high standard of security, performance, reliability, accessibility, scalability and capacity.

Non-Functional Requirement	Justification	KPI
Security		
System must use HTTPS for all data transmissions.	The business uses sensitive customer information such as personal details for their profile and order history, without encrypted data transmission, the sensitive information could be intercepted leading to data breaches. HTTPS ensures a secure communication channel between users and the system thus protecting the business's reputation.	• 100% of the systems pages served over HTTPS. • Security scans must report 0% mixed content warning results.
User authentication data must be securely stored using password hashing.	Passwords in plain text risk a serious security risk, by ensuring hashing methods, the system protects users in the	• 0% of plain text passwords in database.

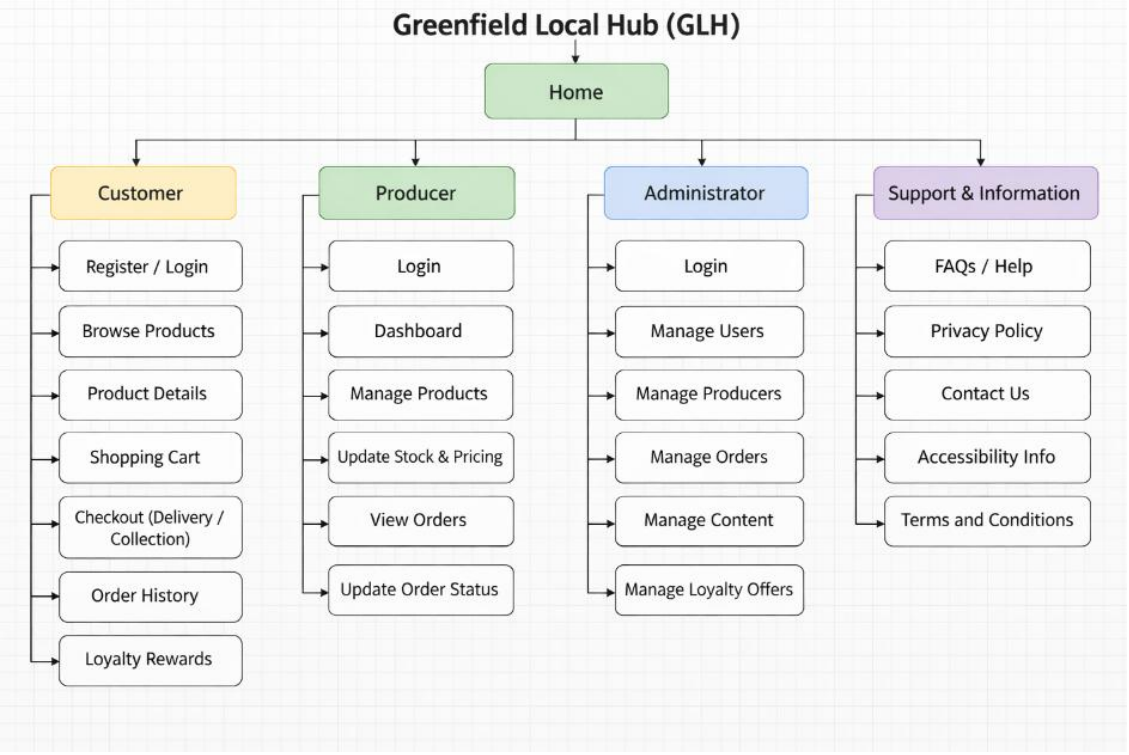
	event of a data breach thus compliance with security best practices while maintaining trust between business, customers and producers.	• 100% of authentication stored using hashing algorithms.
Performance		
The system must load key pages within 2 seconds on a standard 4G or broadband connection.	Customers expect fast loading time for browsing products, viewing producer information and ordering. Slow times cause frustration and can make them leave the page potentially causing revenue loss, thus ensuring fast loading increases likelihood of completed orders.	• Average page load time less than or equal to 2 seconds. • 92% of page load within 2 seconds. • Server response time must remain under 500ms.
The system must support at least 150 concurrent users without performance degradation.	During seasonal peaks or holiday times, users may access the system simultaneously whilst all in communication or ordering from different producers. The system should remain responsive to avoid order failures or delays and ensure consistent service delivery to customers.	• No noticeable latency increases with 150 concurrent users. • Average response time must remain under 1.5 seconds during peak usage. • System CPU must remain below 80% under peak load.
Reliability		
The system must maintain 99% uptime during operational hours.	The systems downtime directly affects the business's way of operating as customers and producers cannot communicate effectively. This directly causes revenue loss and customer frustration. Ensuring uptime ensures consistent service provision and maintains a positive brand image.	• No more than 35 minutes of downtime per month. • Automatic system monitoring must detect outages within 1 minute. • System must restart failed service within 2 minutes automatically.
System must provide error message or contact method, if ordering service becomes unavailable.	In the event of a system failure, customers should not be left wondering or without guidance, providing a clear message or alternative option ensures transparency and preserves customer trust when issues arise.	• Error message and contact method 100% automatically appears to users when system is unavailable or fails. • System must log 100% of system errors in the error monitoring or exception tracking system.
Accessibility		

The system must comply to WCAG 2.1 accessibility standards.	The business aims to serve a diverse range of users including individuals with disabilities. Compliance with accessibility standards ensures equal access whilst preventing legal risks and improves reputation and inclusivity.	• Minimum 95% compliance score in accessibility audits. • Accessibility audit score of 90 or more using automated tools.
All interactive elements must be usable via keyboard only navigation	Users with mobility impairments may not be able to use a mouse, ensuring full keyboard navigation allows these users to browse products and submit orders independently thus preventing exclusion.	• 100% of interactive elements accessible using keyboard controls.
Scalability		
The system must support growth from 1,000 to 10,000 registered users without structural changes.	As the business grows, the platform must scale without requiring redevelopment. A scalable application ensures long term reliability and supports business expansion while maintaining performance.	• Database response times remain under 100ms as user numbers increase.
The system must allow administrators to manage users and system content without developers getting involved.	Allowing administrators to manage this through the system ensures there are no technical bottlenecks and reduces maintenance costs whilst ensuring there is no operational latency. It makes sure that critical updates like suspending fraudulent or malicious accounts occurs in real time including the updating of system content.	• Order status and product updates live within 90 seconds of submission. • 100% of user profiles and site content updates completed by administrators without requiring a code deployment.
Capacity		
The system must support at least 50GB of storage for produce images and required media.	High definition images are essential for helping the customers visualise the produce they may be interested in thus prompt them to make informed decisions. Adequate storage ensures the platform maintains a professional appearance and supports marketing efforts.	• Storage usage remains under 75% capacity at launch. • System must support upload of images up to 10MB each.

The system must retain order records for a minimum of 5 years.	Historical order data is valuable for financial reporting and trend analysis. Retaining these records ensures the business can make informed strategic decisions and meet record keeping obligations.	• 0% data loss for booking records up to 5 years. • Automated database backups must take place every 24 hours.

Hierarchy Diagram

The following diagram illustrates the relationship and hierarchy within the system:



UI(User Interface) Considerations

Visual Design and Branding

The user interface of this solution is designed to reflect a professional, modern but also trustworthy and sustainable service provider for users thus aligning with the expectations of acquiring customers ready to order from producers.

Colours

A clean and professional colour scheme is used consisting of a deep brown for headers and key navigation elements conveying trust and reliability whilst also subconsciously giving the user a naturistic feel as brown is seen throughout nature from trees to soil thus the user subconsciously registers this. This is supported by light green backgrounds to maintain clarity and reduce visual clutter whilst the colour green is proven to make users relax subconsciously. Accent colours such as blue/light grey will be used for order related action (e.g. “Availability” or “Order”) so as to clearly highlight interactive elements whilst making the colour scheme unique to that of competitors giving the business more unique selling points.

Imagery

High quality and high-definition images of the produce will be prioritised to help customers visualise the food they are ordering. This helps in supporting informed decision making for customers and enhances overall professionalism of the digital platform thus increasing customer confidence when making high value purchases. This also assists in trust and transparency between not only the business and customers but producers and customers as well.

Typography

Industry standard Sans-Serif will be used to ensure readability across all devices. Headings will be bold and clear, while text will remain concise and legible ensuring users can quickly understand the information presented to them. Didot will be used on interactive elements to provide unique fonts for the users to see occasionally giving the business more unique selling points to outdo competitors with.

<u>Layout and Navigation</u>

The layout of the system follows a top-down hierarchy structure ensuring intuitive navigation and minimal learning curves for users which is industry standard.

Layout

The interface adopts a responsive mobile first but inclusive design, allowing customers to browse through produce, check availability and manage their orders seamlessly across all their devices. Content will be organised into clear sections to prevent confusion for all the users.

Navigation

A persistent navigation bar will be displayed at the top of the screen thus allowing users to access key sections without excessive scrolling. Order related actions will be clearly grouped to reduce cognitive load on the users and guide users smoothly through the order journey which will in turn increase completion rates as well.

<u>Accessibility</u>

Accessibility is a core consideration within the system interface to ensure the platform can be used by users with a wide range of needs.

Contrast

All text will meet WCAG contrast standards to ensure readability for users with visual impairments whilst keeping font sizes scalable without breaking the layout thus ensuring accessibility across different devices and different user preferences.

Keyboard Navigation and Screen Readers

All interactive elements will be fully accessible using keyboard navigation. Images attached are to have descriptive alt-text to support screen readers thus ensuring that visually impaired users can still understand produce available and order options.

Consistency and Feedback

Consistency

The system will maintain consistent styling and even colour usage and terminology used across all the pages for an organised user experience. This consistency helps users quickly understand how to interact with the system and reduces potential errors.

Feedback

Clear feedback will be provided for user actions like those of confirmation messages after an order has been requested or updating order history. This will reassure users that their actions have been successfully completed and will reduce uncertainty and doubt during ordering process.

Security Considerations and Risk Mitigation

Security is a critical consideration for the business system due to handling of sensitive customer and producer information and data.

Secure authentication: To protect user data, the system will implement secure authentication mechanisms thus ensuring that all customer and producer accounts are protected using encrypted passwords and secure login processes. This would prevent unauthorised access and would protect both the business and users from any data breaches.

Data encryption: All data transmitted between users and Greenfield Local Hub's system will be encrypted using HTTPS thus reducing risk of interception and any data tampering. This would ensure that customers can confidently browse through produce, submit order requests and even manage their orders well without any worry or concern about data security, thus supporting trust and professionalism.

Access Control: Role based access will be applied to ensure that the administrator / manager can only access features that are relevant to their role preventing any accidental or

malicious changes to order data or produce availability whilst ensuring customer data is kept safe and remains protected.

Input Validation: The system will implement input validation across all forms including registration and order request forms. This in turn reduces the risk of malicious input such as an SQL injection or cross-site scripting attacks thus protecting integrity of the system and preventing unauthorised manipulation of order or user data.

Data Backups: Regular data backups will be maintained to protect against data loss caused by potential system failures or security incidents. This ensures that critical order records and customer information can be recovered if required, thus supporting business continuity and protecting Greenfield Local Hub from any operational disruption.

Risk Mitigation Table/ Matrix

Risk Identification	**Probability**	**Impact**	**Mitigation Strategy**
Unauthorised access to customer or producer accounts.	Low	High	Make sure there is secure authentication, encrypted passwords and that access is based upon user role for specific features.
System downtime preventing customer from browsing products or placing orders.	Medium	High	Release the system onto reliable cloud technology with redundancy. It is also beneficial to put in place automatic failover systems to reduce any downtime whenever the system goes down.
Data breach exposing customer information.	Low	High	All sensitive data should be encrypted whilst the business ensures regular security audits takes place. Subsequently, a secure database storage is helpful as well. The business should ensure it complies to GDPR regulations.
Incorrect product availability due to outdated stock levels that the producer did not update.	Medium	Medium	Provide producers with an easy to use and understand dashboard with an automated system for stock validation to reduce outdated listings. System alerts can assist to remind producers to update stocks often.

High system traffic causing slow performance during the peak periods.	Medium	Medium	A scalable cloud infrastructure helps in this, whilst a lot of load balancing helps ensure the system can handle such increases in users without a drop in performance.
Producer errors when entering product information.	Medium	Medium	Adding validation on the form helps with this, by making certain information in the form mandatory and the producer previewing it before it is officially made live.
Producer not using the platform effectively or not adopting the platform.	Medium	Medium	Provide guidance, training materials and guides so that producers understand the simple dashboard tools so that they can manage products and orders received effectively.
Loss of customer trust due to inaccurate product information.	Low	High	Clearly display ordering policies, terms of hire and staff and company information so as to build trust and for customer to know who they are working with whilst ensuring all product listings have clear descriptions, pricing and producer information.
Accessibility not being great thus preventing some users from using the system	Low	Medium	Ensure WCAG are met within the design, keyboard navigation and screen reader support.
Data loss caused by system failure or cyber attack affecting historical ordering records.	Low	High	Maintain secure data storage and daily data backups whilst storing backup copies in separate locations.
Delays in project development due to unclear requirements or scope changes.	Medium	Medium	Regular communication with the client to be made, Project requirements to be documented clearly so as to follow a structured development plan to avoid any scope changes or scope creep.

Legal and Regulatory Requirements

The development of Greenfield Local Hub's digital platform must comply with a range of legal and regulatory requirements including:

Regulation	**Application**
UK General Data Protection Regulation (UK GDPR) & Data Protection Act 2018	The platform collects personal data such as customer names, emails and booking details during their account registration and booking requests. To comply with the act, Greenfield Local Hub must securely store user data and only collect necessary information to deliver the service. Users must also be able to request deletion of their data if required.
Equality Act 2010	The digital platform must be accessible to users with disabilities. A customer using keyboard only navigation must be able to browse produce and producer information, and submit order requests with no problem. Accessibility features such as a colour contrast that is enough and adequate also ensures that no user is unfairly excluded from accessing the service.
Consumer Rights Act 2015	Customers must receive clear and accurate information about the service being provided. This is seen when Rodney Blue browses through products, producer details and submits an order request. The system must clearly display produce and information about the producers clearly and transparently. This ensures services described meet customer expectations and needs.
Electronic Commerce (EC Directive) Regulations 2002	As an online marketplace, the platform must clearly display business information and explain the ordering process. When a customer places an order, the system must clearly confirm that the order has been received and processed. This ensures customers understand and also protects the GLH legally.
Computer Misuse Act 1990	The system must prevent unauthorised access to administrative features. Munya Thomas can access the order history of users but Samantha Riverdale cannot. This ensures ensure compliance and protects business data.
Copyright, Designs and Patents Act 1988	Images, logos and product photographs displayed within the system, must be owned by the business or have permission from producers to be used. This prevents legal issues arising from unauthorised use of digital content and ensures the platform functions properly.

Health and Safety at Work Act 1974	Producers must ensure that food products that they list are safe for consumption and described and shown accurately. The communication should be transparent, information about food and its ingredients must be clearly communicated to users to avoid any health risks.
Privacy and Electronic Communications Regulations	GLH must obtain user consent before sending marketing emails that offer loyalty schemes and rewards for repeat purchasers or top ones. Consent for cookies must also be provided, thus as the system might track user data there is lawful and transparent communication between the business and users.
Environmental Protection Act 1990	Since GLH focuses on locally sourced produce the producers must follow environmentally industry standard practices like waste management and sustainable farming.
Food Safety Act 1990	As the platform connects customers with local producers, it must ensure that producers provide accurate information and they must comply to industry standard food safety standards including things like storage, preparation and the food labelling. This protects GLH from producers selling unsafe and misleading food on the platform.

Compliance with these legal and regulatory requirements ensures that Greenfield Local Hub operates in a manner that adheres to regulations and reduces legal risk for the business whilst strengthening the overall reliability of the program.